KINGSCLIFF HIGH SCHOOL
ACHIEVEMENT THROUGH ENDEAVOUR

YEAR 9 STANDARD

CHAPTER 12

YEAR 9 MATHEMATICS

Probability

5 lessons · three pages each · Easy drill with diagrams, then Medium and worded problems

12.1 Probability

12.2 Experimental probability

12.3 Venn diagrams

12.4 Tree diagrams

12.5 Tables of outcomes

NAME _____ CLASS _____ TEACHER _____

Artwork: a silver jellyfish of spirals near the neck of the left bulb, coloured in purple and silver, part of the Mandelbrot set.

How to use this booklet

1. Each lesson is **three pages**. Read the **learning intention**, copy your teacher's notes into the **Notes** box, and tick each **success criterion** when you can do it.

2. Pages 1 and 2 are **Easy**: small steps, one question at a time. Page 3 is **Medium**, then three **worded problems**.

WE DO **Grey boxes.** Your teacher works each example on the board. **Copy every step** into the space.

YOU DO **Blue boxes.** Your turn. Each question says what to do, step by step. Write each answer on the line that names it, like **P(red) =**.

3. Some Medium questions come from earlier lessons. The tag (like **11.2** or **Year 8**) says which one.

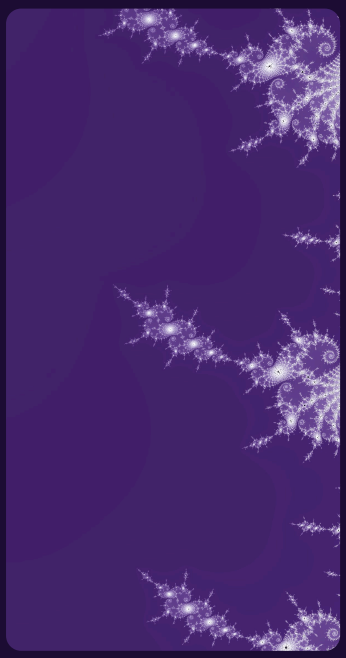
4. Stuck? Read the **Remember** line in the blue box, then look back at your notes and the example. Write probabilities as **fractions in simplest form** unless the question asks for a decimal or a percentage.

Contents and progress tracker

Lesson	Topic	Page	Date	Done
12.1	Probability	3		☐
12.2	Experimental probability	6		☐
12.3	Venn diagrams	9		☐
12.4	Tree diagrams	12		☐
12.5	Tables of outcomes	15		☐

In this chapter you will

- find the probability of an event and of its complement
- find observed (experimental) probabilities and compare them with theoretical probabilities
- use Venn diagrams to find probabilities with "and", "or" and "neither"
- list the outcomes of two-step experiments with tree diagrams
- use tables of outcomes for two dice or spinners
- write probabilities as fractions in simplest form, decimals or percentages



About the artwork

The pictures in this booklet are parts of the **Mandelbrot set**, drawn by a computer that repeats one simple rule, $z \to z^2 + c$, thousands of times for every point. Every booklet shows a different part of the set: this one is a silver jellyfish of spirals near the neck of the left bulb, coloured in purple and silver. Maths can make beautiful things.

Acknowledgement of Country

We acknowledge the Bundjalung people, the Traditional Custodians of the land on which Kingscliff High School stands, and pay our respects to Elders past and present.

12.1 Probability

EASY

Learning intention We are learning to find the probability of an event and of its complement.

Success criteria
- ☐ I can list the outcomes and count the total
- ☐ I can write a probability as a fraction in simplest form
- ☐ I can find P(not A) = 1 − P(A)

NOTES

Copy what your teacher writes on the board.

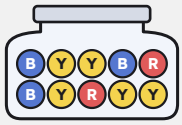
Date _____

WE DO

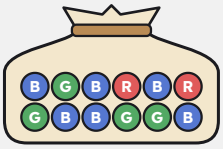
With your teacher: copy what they write on the board.

Example 1 Find each probability as a fraction in simplest form.

a P(red), P(not red)



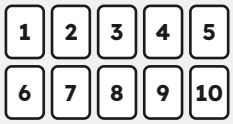
b P(blue), P(not blue)



c P(A), P(not A)



d P(more than 6)



YOU DO

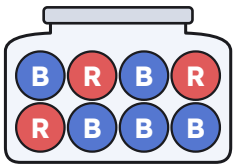
On your own: work down the box, one question at a time. Show your working.

1 QUESTION

REMEMBER P(event) = number in the event ÷ total · simplify: divide the top and bottom by the same number · P(not event) = 1 − P(event)

QUESTION 1 Count the total, find P(colour) in simplest form, then P(not colour) = 1 − P(colour).

a

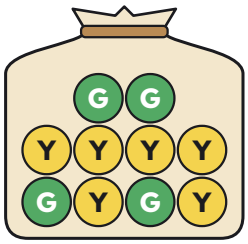


Total = _____

P(red) = _____

P(not red) = _____

b



Total = _____

P(green) = _____

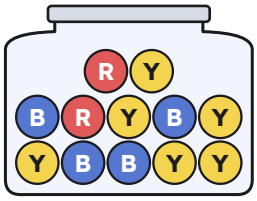
P(not green) = _____

YOU DO On your own: keep going, one question at a time. Show your working. 3 QUESTIONS

REMEMBER Count every counter or card for the total · simplify the fraction · P(colour) + P(not colour) = 1

QUESTION 2 Count the total, find P(colour) in simplest form, then P(not colour) = 1 − P(colour).

(a)

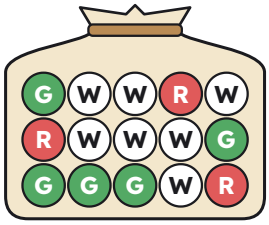


Total = _____

P(blue) = _____

P(not blue) = _____

(b)



Total = _____

P(white) = _____

P(not white) = _____

QUESTION 3 Count the cards, find P(letter) in simplest form, then P(not letter) = 1 − P(letter).

(a)

P E P P E R

Total = _____

P(E) = _____

P(not E) = _____

(b)

L O L L I P O P

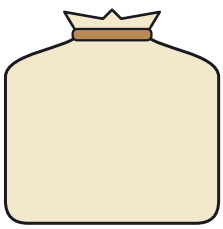
Total = _____

P(L) = _____

P(not L) = _____

QUESTION 4 Draw the counters in the bag or jar. Then find P(colour) and P(not colour).

(a) A bag holds 3 red, 5 blue and 4 green counters.



Total = _____

P(green) = _____

P(not green) = _____

(b) A jar holds 6 yellow, 2 red and 8 blue counters.



Total = _____

P(red) = _____

P(not red) = _____

WE DO With your teacher: copy what they write on the board.

Example 2 **Find each probability as a fraction in simplest form.**

a A jar holds only red and yellow marbles. P(red) = 40%. Find P(yellow).

b The jar in part a holds 40 marbles. How many are red?

c A card is drawn from a standard deck. Find P(red card).



4 suits × 13 = 52

d Find P(King or Queen).



4 suits × 13 = 52

YOU DO On your own: work in order. A tag shows an earlier lesson. Show your working. 6 QUESTIONS

REMEMBER A standard deck: 52 cards · hearts and diamonds are red · each suit has A, 2 to 10, J, Q, K · P(not A) = 1 – P(A)

5 Find P(heart) and P(not heart).



4 suits × 13 = 52

P(heart) = _____

P(not heart) = _____

6 Write 18/24 in simplest form. Year 8

Answer _____

7 Only red and yellow. P(red) = 3/8.

Working _____

P(yellow) = _____

8 24 counters. P(red) = 1/4. How many red?

Working _____

Red counters = _____

9 Find the gradient through (1, 2) and (3, 8). 11.3

Answer _____

10 A letter of MATHEMATICS is picked.

Total = _____

P(M) = _____

P(not M) = _____

WORDED PROBLEMS Draw a diagram if it helps, then solve. Show your working. 3 PROBLEMS

1 A class has 12 boys and 18 girls. One student is chosen at random to read. Find the probability that the student is a girl, and that the student is not a girl.

Answer _____

2 A bag holds only red, blue and green counters. P(red) = 3/10 and P(blue) = 9/20. Find P(green). The bag holds 40 counters: how many are green?

Answer _____

3 A raffle sells 500 tickets and Mia buys 8 of them. Find the probability that Mia wins first prize, and that she does not.

Answer _____

12.2 Experimental probability

EASY

Learning intention We are learning to find observed probabilities and compare them with theoretical ones.

Success criteria
- ☐ I can add the frequencies to find the number of trials
- ☐ I can find the observed and the theoretical probability
- ☐ I can find the expected number of times an outcome happens

NOTES Copy what your teacher writes on the board. **Date** _____

WE DO With your teacher: copy what they write on the board.

Example 1 **Find each probability as a fraction in simplest form.**

a Coin: observed and theoretical P(head)

Outcome	H	T
Frequency	28	22

b 40 spins: 3 came up 12 times. P(3)?



c Equal spinner R, B, G: P(red)

Outcome	R	B	G
Frequency	12	9	9

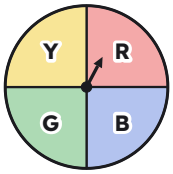
d Expected frequency: the spinner in part b is spun 200 times. How many 3s do you expect?

YOU DO On your own: work down the box, one question at a time. Show your working. 1 QUESTION

REMEMBER Observed P = frequency ÷ total number of trials · theoretical P = outcomes in the event ÷ equally likely outcomes

QUESTION 1 Add the frequencies. Observed P = frequency ÷ total; theoretical P from the spinner.

a



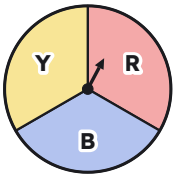
Outcome	R	B	G	Y
Frequency	13	9	8	10

Total = _____

Observed P(red) = _____

Theoretical P(red) = _____

b



Outcome	R	B	Y
Frequency	18	24	18

Total = _____

Observed P(blue) = _____

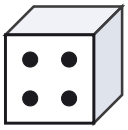
Theoretical P(blue) = _____

YOU DO On your own: keep going, one question at a time. Show your working. 3 QUESTIONS

REMEMBER The total is all the frequencies added · a die has 6 equally likely outcomes · a sector twice as big counts as 2 equal parts

QUESTION 2 Add the frequencies. Observed P = frequency ÷ total; theoretical P from the die.

(a)



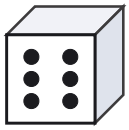
Outcome	1	2	3	4	5	6
Frequency	8	11	9	12	10	10

Total = _____

Observed P(4) = _____

Theoretical P(4) = _____

(b)



Outcome	1	2	3	4	5	6
Frequency	18	22	19	21	16	24

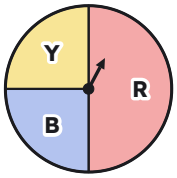
Total = _____

Observed P(even) = _____

Theoretical P(even) = _____

QUESTION 3 As decimals: observed P = frequency ÷ total; theoretical P from the sector sizes.

(a)



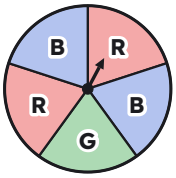
Outcome	R	B	Y
Frequency	46	27	27

Total = _____

Observed P(red) = _____

Theoretical P(red) = _____

(b)



Outcome	R	B	G
Frequency	18	22	10

Total = _____

Observed P(blue) = _____

Theoretical P(blue) = _____

QUESTION 4 Tally the results to complete the table. Then find the observed and theoretical P.

(a) A coin was tossed 20 times: H T T H H T H H H T T H T H H T H H T H

Outcome	H	T
Tally		
Frequency		

Total = _____

Observed P(head) = _____

Theoretical P(head) = _____

(b) An equal R, B, G spinner gave: R B B G R B R G B B R B G B R

Outcome	R	B	G
Tally			
Frequency			

Total = _____

Observed P(blue) = _____

Theoretical P(blue) = _____

12.2 Experimental probability

MEDIUM

WE DO With your teacher: copy what they write on the board.

Example 2 **The red sector is twice the size of each other sector. 200 spins gave these results. Answer as decimals.**

a Observed P(red)

Outcome	Frequency
R	72
B	29
G	35
Y	30
P	34

b Observed P(blue or green)

c Theoretical P(red), P(not purple)



d Expected number of reds in 500 spins

YOU DO On your own: work in order. A tag shows an earlier lesson. Show your working. 6 QUESTIONS

REMEMBER Count a double sector as two equal parts · expected frequency = theoretical P × number of trials · decimal = top ÷ bottom

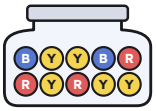
5 Find the observed P(2) from 50 spins.

Outcome	1	2	3	4
Frequency	7	14	12	17

Total = _____

Observed P(2) = _____

6 Find P(blue) and P(not blue). 12.1



P(blue) = _____

P(not blue) = _____

7 P(red) = 0.35. Reds expected in 400 spins?

Working _____

Expected reds = _____

8 A die gave 180 sixes in 600 rolls.

Expected 6s = _____

Fair? Why? _____

9 Write 7/20 as a decimal and a percentage. Year 8

Answer _____

10 Find the observed P(red) as a percentage.

Outcome	R	B	G
Frequency	18	12	10

Total = _____

Observed P(red) = _____

WORDED PROBLEMS Draw a diagram if it helps, then solve. Show your working. 3 PROBLEMS

1 A basketballer made 36 of her last 48 free throws. Find the experimental probability that she makes her next throw. How many of her next 200 throws would you expect her to make?

Answer _____

2 A die is rolled 300 times and lands on 6 a total of 71 times. Find the observed and the theoretical probability of a 6, as decimals to two decimal places. Do you think the die is fair?

Answer _____

3 Of 120 cars that passed the school gate, 18 were red. Estimate the probability that the next car is red. How many of the next 400 cars would you expect to be red?

Answer _____

12.3 Venn diagrams

EASY

Learning intention We are learning to use Venn diagrams to find probabilities.

Success criteria
- ☐ I can read the number in each region of a Venn diagram
- ☐ I can find P(A and B), P(A or B) and P(neither)
- ☐ I can complete a Venn diagram from given information

NOTES

Copy what your teacher writes on the board.

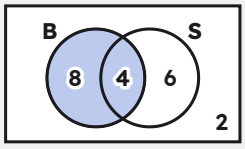
Date _____

WE DO

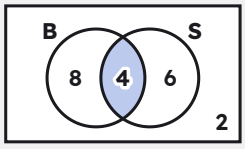
With your teacher: copy what they write on the board.

Example 1 B = plays basketball, S = swims. Find each probability in simplest form.

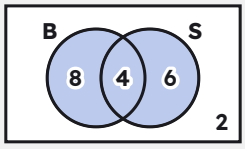
a P(B)



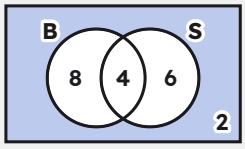
b P(B and S)



c P(B or S)



d P(neither)



YOU DO

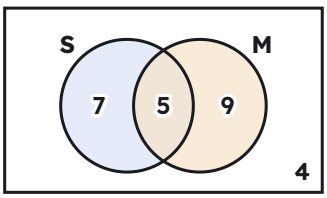
On your own: work down the box, one question at a time. Show your working.

1 QUESTION

REMEMBER Total = all four numbers · A = A only + both · A or B = the three numbers in the circles · neither = the number outside

QUESTION 1 Add all four numbers for the total. Then add the regions for each event.

a Sport (S), music (M)



Total = _____

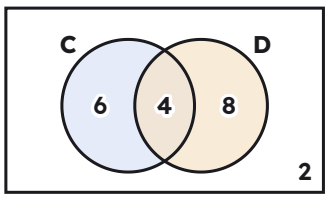
P(S) = _____

P(S and M) = _____

P(S or M) = _____

P(neither) = _____

b Has a cat (C), has a dog (D)



Total = _____

P(C) = _____

P(C and D) = _____

P(C or D) = _____

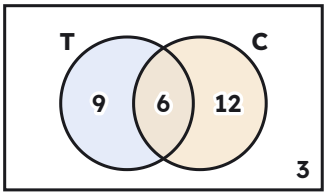
P(neither) = _____

YOU DO On your own: keep going, one question at a time. Show your working. 3 QUESTIONS

REMEMBER "and" is the overlap only · "or" is everything inside the circles · both = A + B + neither − total

QUESTION 2 Add all four numbers for the total. Then add the regions for each event.

(a) Drinks tea (T), drinks coffee (C)



Total = _____

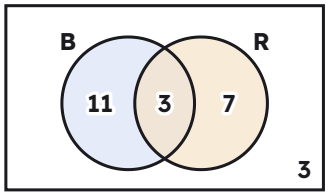
P(T) = _____

P(T and C) = _____

P(T or C) = _____

P(neither) = _____

(b) Catches a bus (B), a train (R)



Total = _____

P(B) = _____

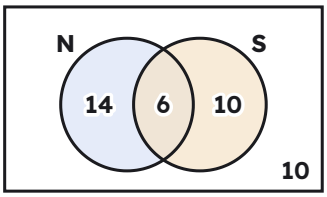
P(B and R) = _____

P(B or R) = _____

P(neither) = _____

QUESTION 3 Add all four numbers for the total. Then add the regions for each event.

(a) Plays netball (N), soccer (S)



Total = _____

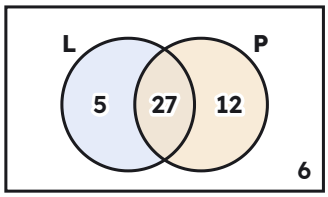
P(N) = _____

P(N and S) = _____

P(N or S) = _____

P(neither) = _____

(b) Owns a laptop (L), a phone (P)



Total = _____

P(L) = _____

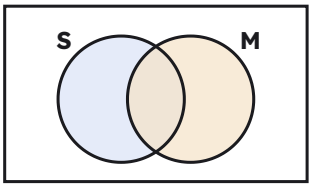
P(L and P) = _____

P(L or P) = _____

P(neither) = _____

QUESTION 4 Both = A + B + neither − total. Fill in the Venn diagram, then each probability.

(a) 32 students: 20 play sport (S), 15 play music (M), 5 do neither.



Both = _____

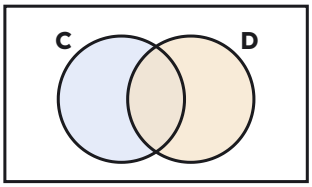
P(S) = _____

P(S and M) = _____

P(S or M) = _____

P(neither) = _____

(b) 40 people: 25 have a cat (C), 18 have a dog (D), 6 have neither.



Both = _____

P(C) = _____

P(C and D) = _____

P(C or D) = _____

P(neither) = _____

12.3 Venn diagrams

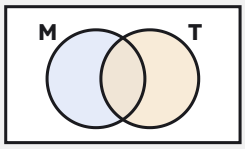
MEDIUM

WE DO With your teacher: copy what they write on the board.

Example 2 **30 teenagers were surveyed: 21 like music (M), 18 like TV (T) and 4 like neither.**

a How many like both music and TV?

b Complete the Venn diagram.



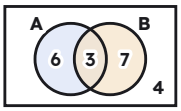
c P(music but not TV)

d P(music or TV, but not both)

YOU DO On your own: work in order. A tag shows an earlier lesson. Show your working. 6 QUESTIONS

REMEMBER Both = A + B + neither − total · A only = A − both · "not B" is everything outside circle B

5 Find P(A only) and P(not B).



P(A only) = _____

P(not B) = _____

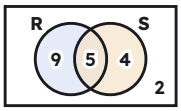
6 Find the observed P(tail). 12.2

Outcome	H	T
Frequency	14	26

Total = _____

Observed P(T) = _____

7 Find P(only one of R and S).



Regions _____

P(only one) = _____

8 24 students: 15 band, 12 choir, 5 neither.

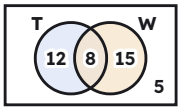
Both = _____

P(both) = _____

9 Find P(not a club) for a standard deck. 12.1

Answer _____

10 Find P(neither) and P(T or W).



P(neither) = _____

P(T or W) = _____

WORDED PROBLEMS Draw a diagram if it helps, then solve. Show your working. 3 PROBLEMS

1 In a class of 28 students, 15 play netball, 10 play soccer and 6 play both. Draw a Venn diagram. Find the probability that a student chosen at random plays neither sport.

Answer _____

2 Of 50 shoppers, 32 bought bread, 20 bought milk and 8 bought neither. How many bought both? Find the probability that a shopper chosen at random bought bread or milk.

Answer _____

3 A whole number from 1 to 20 is chosen at random. E = even numbers and T = multiples of 3. Draw the Venn diagram, then find P(E and T) and P(E or T).

Answer _____

12.4 Tree diagrams

EASY

Learning intention We are learning to list the outcomes of two-step experiments with tree diagrams.

Success criteria
- ☐ I can draw a tree diagram for two steps
- ☐ I can list every outcome and count them
- ☐ I can find probabilities, with or without replacement

NOTES

Copy what your teacher writes on the board.

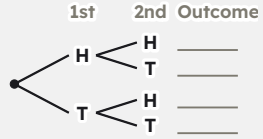
Date _____

WE DO

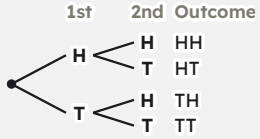
With your teacher: copy what they write on the board.

Example 1 Use the tree diagram. Write probabilities in simplest form.

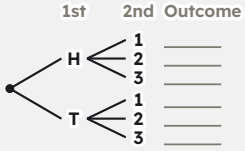
a Two coins: list the outcomes



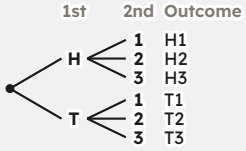
b Two coins: P(two tails)



c Coin, then spinner 1–3: list them



d P(tail and an odd number)



YOU DO

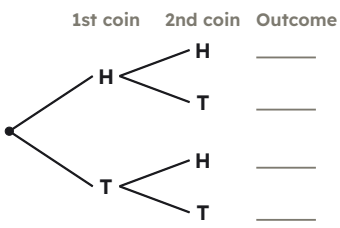
On your own: work down the box, one question at a time. Show your working.

1 QUESTION

REMEMBER Follow each path from the start to a branch end: that is one outcome · P(event) = outcomes in the event ÷ all outcomes

QUESTION 1 Write each outcome at the end of its branch. Count them, then find each probability.

a Two coins are tossed.

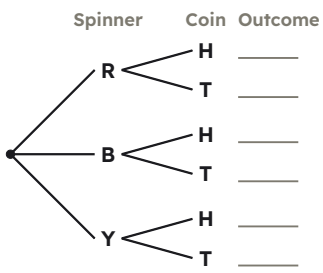


Number of outcomes = _____

P(two heads) = _____

P(at least one head) = _____

b A spinner (R, B, Y) is spun, then a coin is tossed.



Number of outcomes = _____

P(red and head) = _____

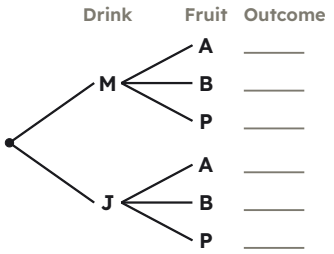
P(a head) = _____

YOU DO On your own: keep going, one question at a time. Show your working. 3 QUESTIONS

REMEMBER Outcomes = 1st step × 2nd step · without replacement, the 2nd step has one less · "at least one" means one or more

QUESTION 2 Write each outcome at the end of its branch. Count them, then find each probability.

(a) Drink: milk (M) or juice (J). Fruit: apple (A), banana (B) or pear (P).

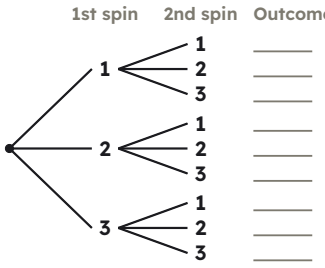


Number of outcomes = _____

P(juice and pear) = _____

P(banana) = _____

(b) A spinner numbered 1, 2, 3 is spun twice.



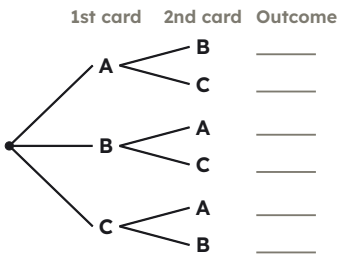
Number of outcomes = _____

P(a double) = _____

P(a total of 5) = _____

QUESTION 3 Not put back: the 2nd step has one less. Write each outcome, then each probability.

(a) Two cards are drawn from A, B and C without replacement.

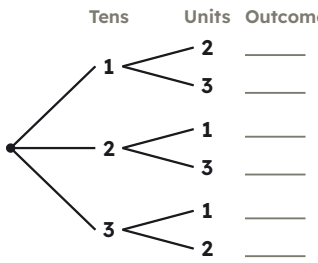


Number of outcomes = _____

P(A then B) = _____

P(no C) = _____

(b) Two of the cards 1, 2, 3 are drawn, not replaced, to make a 2-digit number.



Number of outcomes = _____

P(an even number) = _____

P(more than 20) = _____

QUESTION 4 Draw the tree diagram and list the outcomes. Then find each probability.

(a) A coin is tossed, then a spinner with 1, 2, 3, 4 is spun.

Coin Spinner Outcome

Number of outcomes = _____

P(head and 4) = _____

P(an even number) = _____

(b) Choose pasta (P) or salad (S), then water (W), juice (J) or milk (M).

Main Drink Outcome

Number of outcomes = _____

P(salad and milk) = _____

P(not water) = _____

12.4 Tree diagrams

MEDIUM

WE DO With your teacher: copy what they write on the board.

Example 2 **Cards A, B, C and D are in a bag. Two are drawn at random without replacement.**

a How many outcomes are there? List them.

b P(the letters A and B)

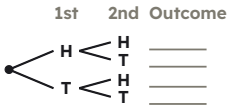
c P(A or B or both)

d P(no letter D)

YOU DO On your own: work in order. A tag shows an earlier lesson. Show your working. 6 QUESTIONS

REMEMBER Without replacement: 4 cards, then 3, so 4 × 3 = 12 outcomes · AB and BA are different outcomes

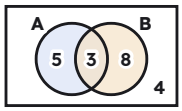
5 Two coins: find P(one of each).



Outcomes = _____

P(HT or TH) = _____

6 Find P(A and B). 12.3



Total = _____

P(A and B) = _____

7 How many outcomes: a die, then a coin?

Working _____

Outcomes = _____

8 A 1–4 spinner is spun twice: P(double)?

Outcomes = _____

P(double) = _____

9 P(red) = 0.3. Reds expected in 250 spins? 12.2

Answer _____

10 Two of X, Y, Z, not replaced: P(X first)?

Outcomes = _____

P(X first) = _____

WORDED PROBLEMS Draw a diagram if it helps, then solve. Show your working. 3 PROBLEMS

1 A family has two children. Draw a tree diagram for boy (B) or girl (G), with each equally likely. Find the probability that the family has one boy and one girl.

Answer _____

2 Sam chooses a T-shirt (red R, blue B or green G) and a cap (black K or white W) at random. Draw a tree diagram. Find the probability of a blue T-shirt with a white cap, and of no red.

Answer _____

3 Ali (A), Ben (B) and Cara (C) put their names in a hat. Two names are drawn without replacement: captain, then vice-captain. List the outcomes and find the probability that Cara is chosen.

Answer _____

12.5 Tables of outcomes

EASY

Learning intention We are learning to use tables of outcomes to find probabilities for two dice or spinners.

Success criteria
- ☐ I can complete a table of outcomes
- ☐ I can count the cells for an event
- ☐ I can find probabilities with "or" and "at least"

NOTES

Copy what your teacher writes on the board.

Date _____

WE DO

With your teacher: copy what they write on the board.

Example 1 Use each table of outcomes. Write probabilities in simplest form.

a Two 1–3 spinners, added: complete it

2nd \ 1st	1	2	3
1	2	3	4
2	3		
3	4		

b P(sum of 4)

2nd \ 1st	1	2	3
1	2	3	4
2	3	4	5
3	4	5	6

c P(odd sum)

2nd \ 1st	1	2	3
1	2	3	4
2	3	4	5
3	4	5	6

d Coin and die: P(head and 3)

	1	2	3	4	5	6
H	H1	H2	H3	H4	H5	H6
T	T1	T2	T3	T4	T5	T6

YOU DO

On your own: work down the box, one question at a time. Show your working.

1 QUESTION

REMEMBER Each cell is one outcome · total cells = rows × columns · P(event) = cells in the event ÷ total cells

QUESTION 1 Complete the table. Then count the cells for each event, out of the total.

(a)

Spinner 1 / Spinner 2

+	1	2	3
1	2	3	4
2	3		
3	4		
4	5		

Total = _____

P(sum of 5) = _____

P(even sum) = _____

P(sum more than 5) = _____

(b)

Die / Coin

	1	2	3	4	5	6
H	H1	H2	H3	H4	H5	H6
T	T1					

Total = _____

P(head and even) = _____

P(a 6) = _____

P(tail or 1) = _____

YOU DO On your own: keep going, one question at a time. Show your working. 3 QUESTIONS

REMEMBER Fill in each cell from the top number and the side number · "at least 2" means 2 or more · "or": count the cells of both

QUESTION 2 Complete the table. Then count the cells for each event, out of the total.

(a)

Die 1

Die 2 \ +	1	2	3	4	5	6
1	2	3	4	5	6	7
2	3					
3	4					
4	5					
5	6					
6	7					

Total = _____

P(sum of 8) = _____

P(sum of 10 or more) = _____

P(even sum) = _____

(b)

Die 1

Die 2 \ ×	1	2	3	4	5	6
1	1	2	3	4	5	6
2	2					
3	3					
4	4					
5	5					
6	6					

Total = _____

P(product of 12) = _____

P(odd product) = _____

P(product more than 20) = _____

QUESTION 3 Complete the table. Then count the cells for each event, out of the total.

(a)

Spinner 1

Spinner 2	R	B	G
R	RR	BR	GR
B	RB		

Total = _____

P(same colour) = _____

P(at least one blue) = _____

P(no red) = _____

(b)

Spinner 1

Spinner 2 \ ×	1	2	3	4
1	1	2	3	4
2	2			
3	3			
4	4			

Total = _____

P(product of 4) = _____

P(odd product) = _____

P(product more than 6) = _____

QUESTION 4 Fill in every cell of the table yourself. Then find each probability.

(a) Two spinners, each numbered 1, 2, 3, are spun and the numbers added.

Spinner 1

Spinner 2 \ +	1	2	3
1			
2			
3			

Total = _____

P(sum of 4) = _____

P(sum more than 4) = _____

P(odd sum) = _____

(b) Spinners 1–4 and 1–3 are spun. Find the difference of the numbers.

Spinner 1

Spinner 2 \ −	1	2	3	4
1				
2				
3				

Total = _____

P(difference of 0) = _____

P(difference of at least 2) = _____

P(odd difference) = _____

12.5 Tables of outcomes

MEDIUM

WE DO With your teacher: copy what they write on the board.

Example 2 **Two dice are rolled and the difference between the numbers is found.**

a Complete the table of outcomes.

–	1	2	3	4	5	6
1						
2						
3						
4						
5						
6						

b P(difference of 0)

c P(difference of at least 2)

d P(difference of 4 or 5)

YOU DO On your own: work in order. A tag shows an earlier lesson. Show your working. 6 QUESTIONS

REMEMBER Difference = larger − smaller · two dice give 6 × 6 = 36 outcomes · count the cells, then simplify

5 Two coins: complete, then P(one of each).

	H	T
H		
T		

Total = _____

P(HT or TH) = _____

6 How many outcomes: 3 shirts, then 4 hats? 12.4

Answer _____

7 Two dice are rolled: find P(sum of 9).

Total = _____

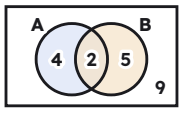
P(sum of 9) = _____

8 Two dice are rolled: find P(both even).

Total = _____

P(both even) = _____

9 Find P(A or B). 12.3



Total = _____

P(A or B) = _____

10 Two 1–3 spinners: P(product is even)?

Total = _____

P(even product) = _____

WORDED PROBLEMS Draw a diagram if it helps, then solve. Show your working. 3 PROBLEMS

1 In a game, two dice are rolled and the numbers are added. You win if the sum is 7 or 11. Find the probability of winning.

Answer _____

2 Two spinners, one numbered 1 to 4 and one numbered 1 to 5, are spun and the numbers multiplied. Draw a table of outcomes and find the probability that the product is more than 10.

Answer _____

3 Mia rolls two dice and records the larger number (or the number, if both are the same). Find the probability that the larger number is 6, and the probability that it is 1.

Answer _____